CSA13 – CLOUD COMPUTING AND BIG DATA ANALYTICS

LAB RECORD – LIST OF EXPERIMENTS

Ex. No: 1 Creation of Virtual Machine using VMware Workstation

Aim

To demonstrate virtualization by installing a Type-2 Hypervisor, VMware Workstation, and creating a Virtual Machine with the specified CPU, RAM, and storage using a Windows/Linux host operating system.

PROCEDURE

Step 1: Install VMware Workstation on the host computer.

Step 2: Open VMware Workstation.

Step 3: Click Create a New Virtual Machine.

Step 4: Select Typical (Recommended) configuration**.**

Step 5: Select the required Windows/Linux ISO image.

Step 6: Enter the Virtual Machine name and select the storage location.

Step 7: Configure the processor according to the given requirement.

Step 8: Allocate the required RAM to the Virtual Machine.

Step 9: Create a virtual hard disk with the required storage capacity.

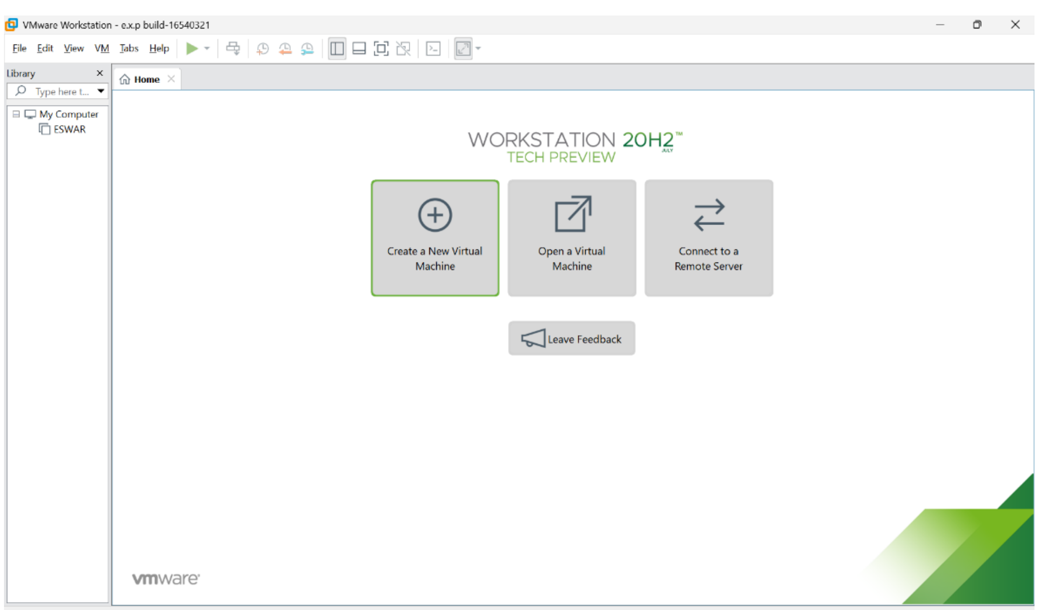
Step 10: Click Finish to create the Virtual Machine.

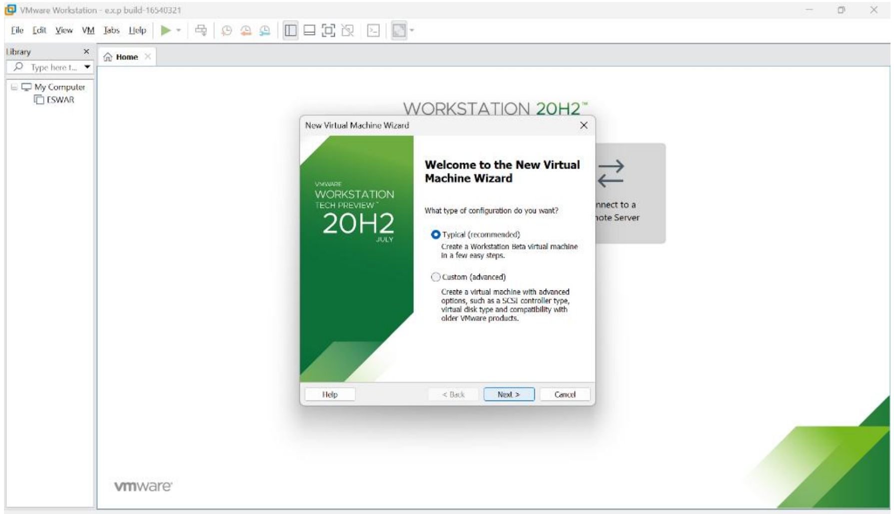
Step 11: Power on the Virtual Machine and install the selected operating system.

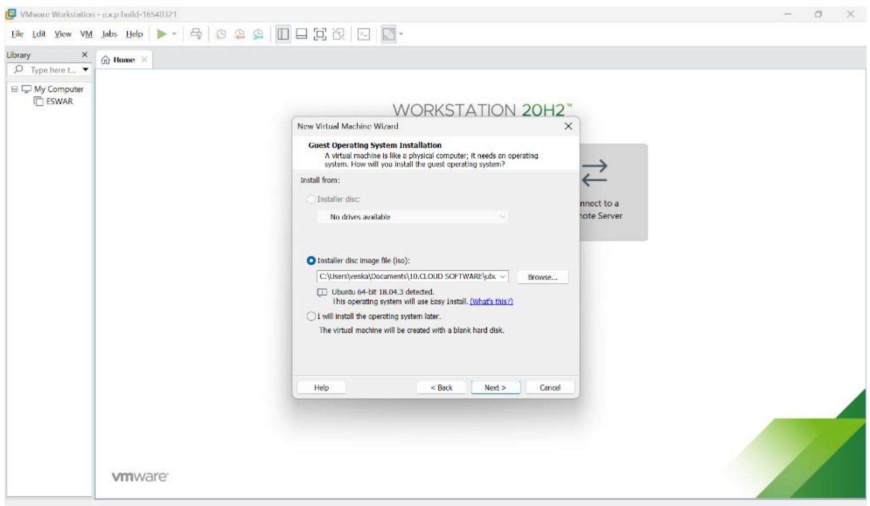
Step 12: Verify the CPU, RAM, and storage configuration.

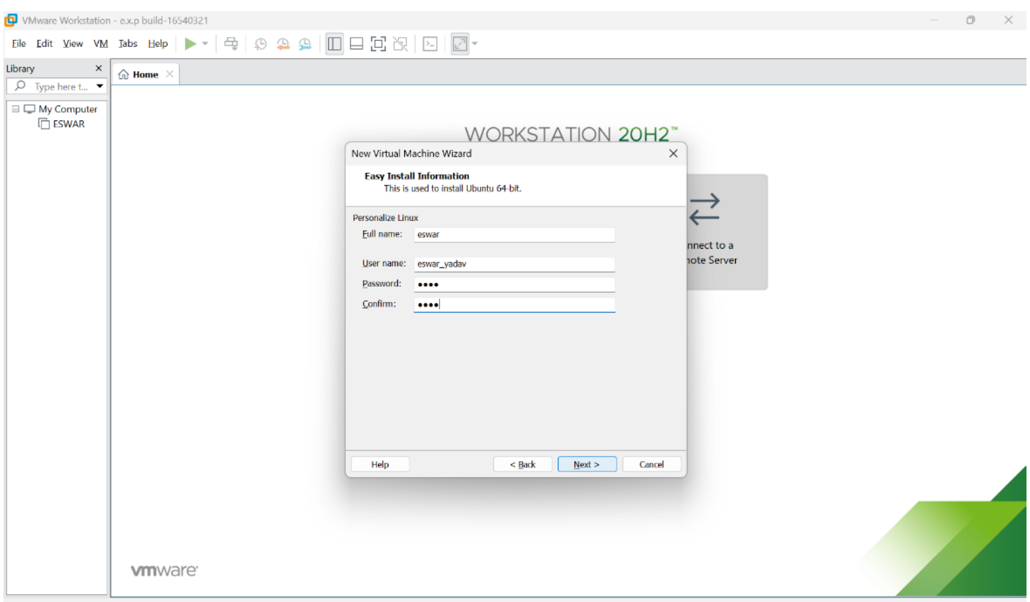
Step 13: The Virtual Machine has been successfully created using VMware Workstation.

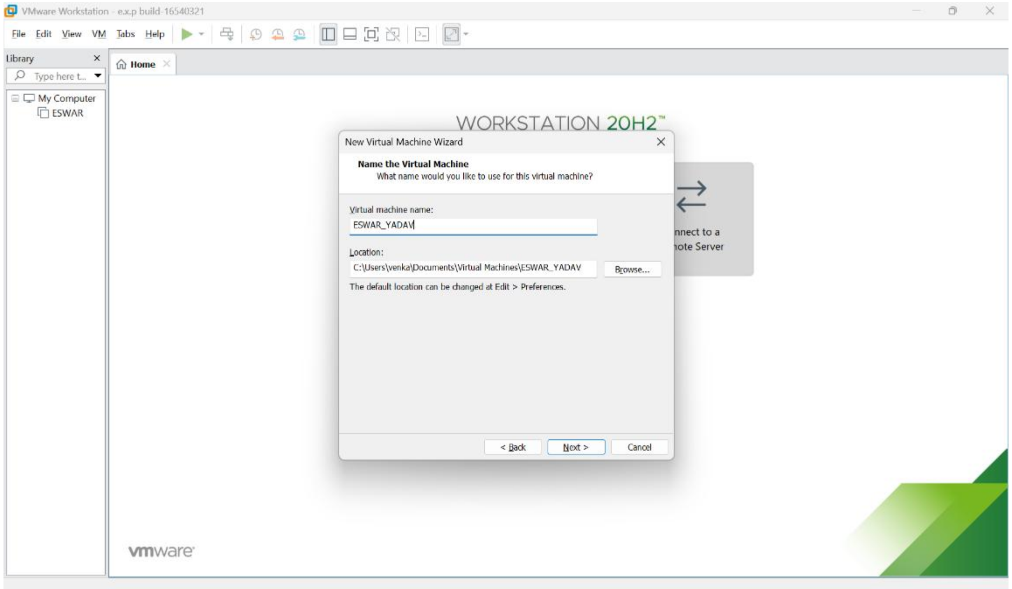
Output




VMware Workstation - e.x.p build-16540321
New Virtual Machine Wizard
WORKSTATION 20H2
Welcome to the New Virtual Machine Wizard
What type of configuration do you want?
Typical (recommended)
Create a Workstation Beta virtual machine in a few easy steps.
Custom (advanced)
Create a virtual machine with advanced options, such as a SCSI controller type, virtual disk type and compatibility with older VMware products.
Help
< Back
Next >
Cancel


VMware Workstation - e.x.p build-16540321
New Virtual Machine Wizard
Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?
Install from:
Installer disc:
No drives available
Installer disc image file (iso):
Browse...
Ubuntu 64-bit 18.04.3 detected.
This operating system will use Easy Install. (What's this?)
I will install the operating system later.
The virtual machine will be created with a blank hard disk.
Help
< Back
Next >
Cancel


VMware Workstation - e.x.p build-16540321
New Virtual Machine Wizard
Easy Install Information
This is used to install Ubuntu 64-bit.
Personalize Linux
Full name: eswar
User name: eswar_yadav
Password:
Confirm:
Help
< Back
Next >
Cancel



Result

Thus the Virtual Machine was successfully created using **VMware Workstation**, a Type-2 Hypervisor, with the specified CPU, RAM, and storage configuration.

Ex. No: 2 Creation of Virtual Machine using VMware Workstation

Set 2: 3 CPU, 10 GB RAM, 10 GB Storage

Aim

To demonstrate virtualization by installing a Type-2 Hypervisor, VMware Workstation, and creating a Virtual Machine with the specified CPU, RAM, and storage using a Windows/Linux host operating system.

PROCEDURE

Step 1: Install VMware Workstation on the host computer.

Step 2: Open VMware Workstation.

Step 3: Click Create a New Virtual Machine.

Step 4: Select Typical (Recommended) configuration**.**

Step 5: Select the required Windows/Linux ISO image.

Step 6: Enter the Virtual Machine name and select the storage location.

Step 7: Configure the processor according to the given requirement.

Step 8: Allocate the required RAM to the Virtual Machine.

Step 9: Create a virtual hard disk with the required storage capacity.

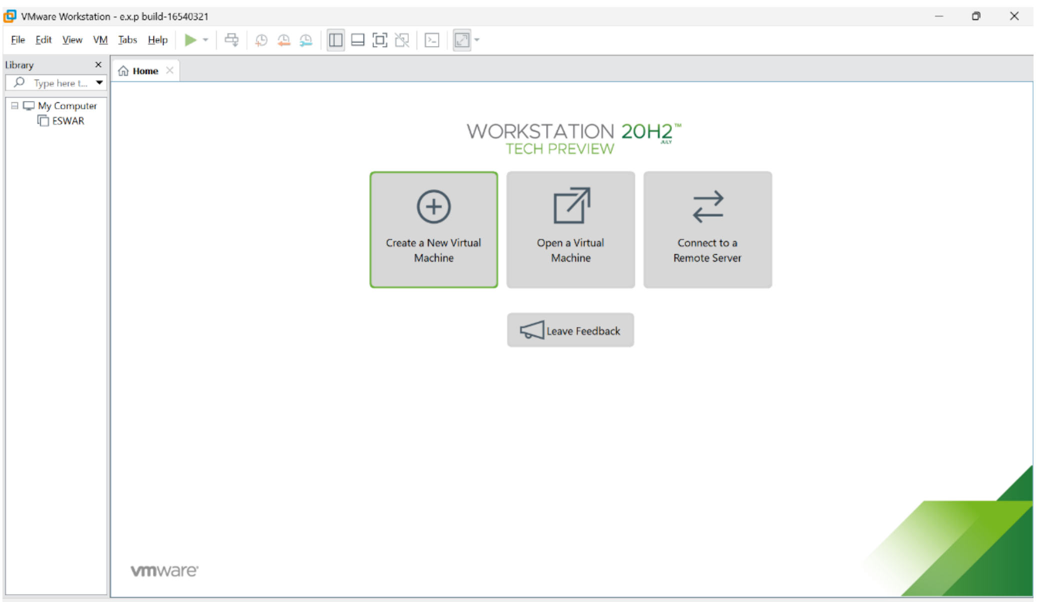
Step 10: Click Finish to create the Virtual Machine.

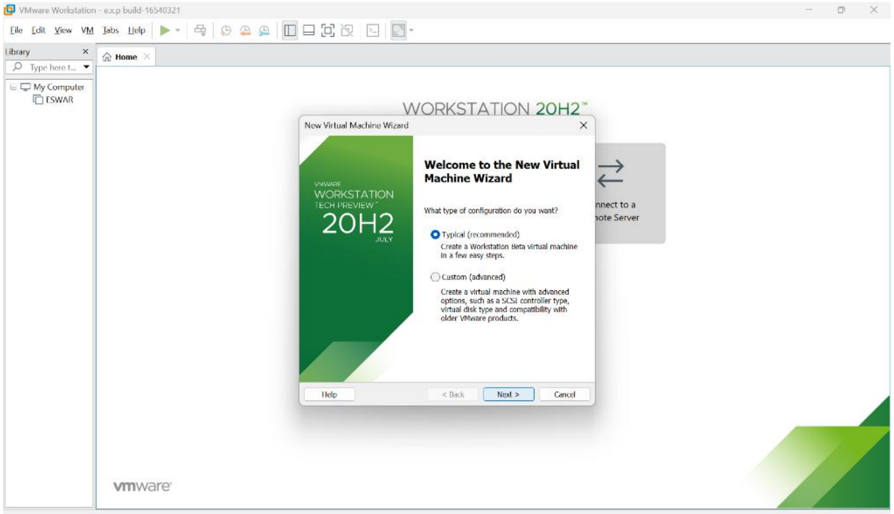
Step 11: Power on the Virtual Machine and install the selected operating system.

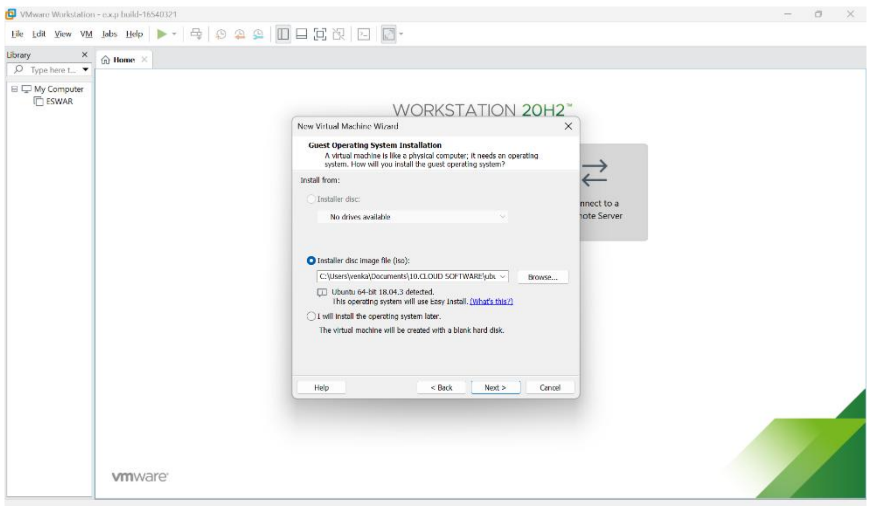
Step 12: Verify the CPU, RAM, and storage configuration.

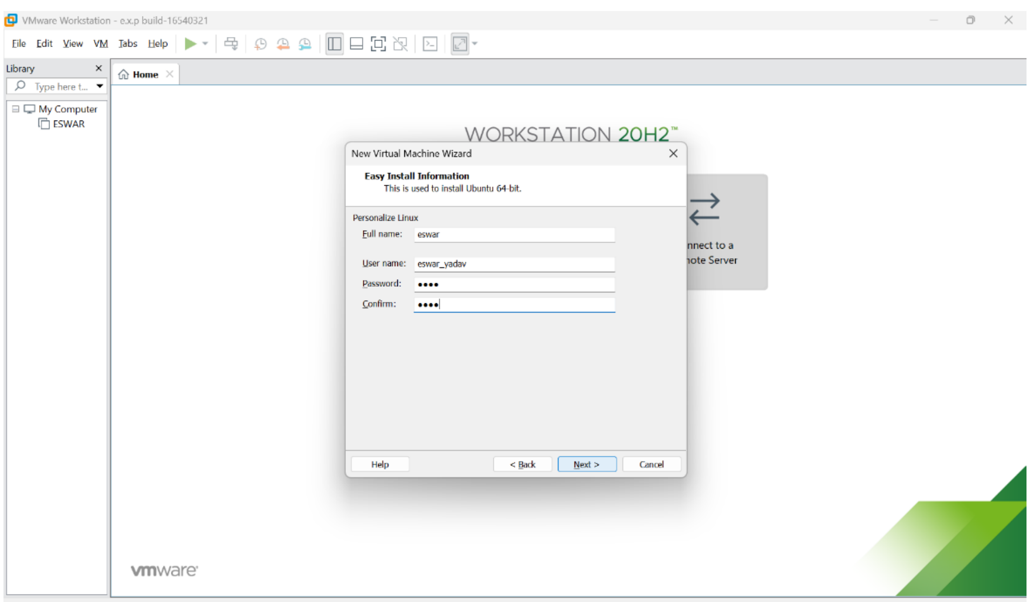
Step 13: The Virtual Machine has been successfully created using VMware Workstation.

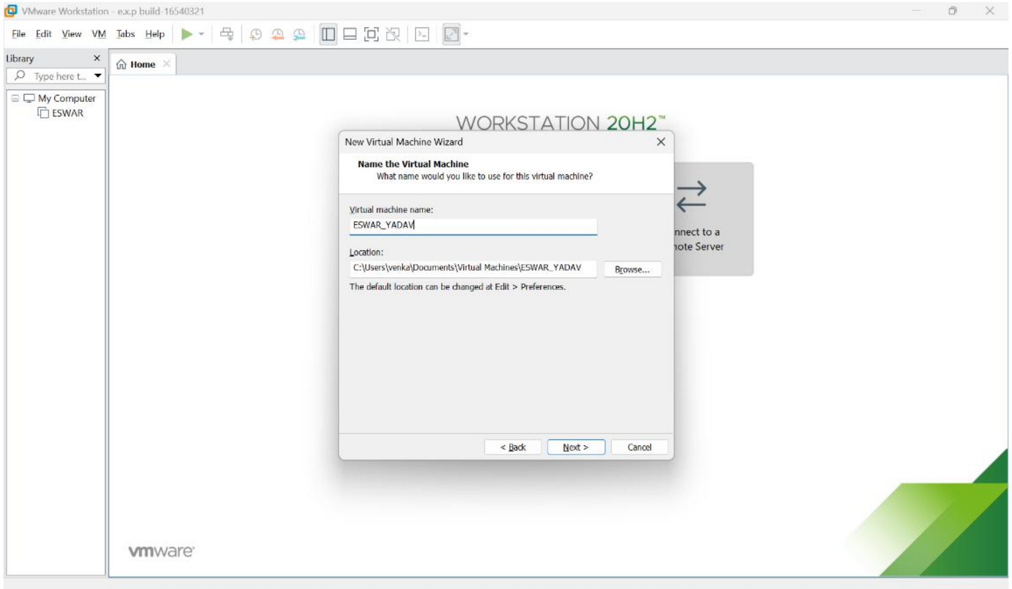
Output




VMware Workstation - e.x.p build-16540321
WORKSTATION 20H2
New Virtual Machine Wizard
Welcome to the New Virtual Machine Wizard
What type of configuration do you want?
Typical (recommended)
Create a Workstation Beta virtual machine in a few easy steps.
Custom (advanced)
Create a virtual machine with advanced options, such as a SCSI controller type, virtual disk type and compatibility with older VMware products.
Help
< Back
Next >
Cancel


VMware Workstation - e.x.p build-16540321
WORKSTATION 20H2
New Virtual Machine Wizard
Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?
Install from:
Installer disc:
No drives available
Installer disc image file (iso):
Browse...
Ubuntu 64-bit 18.04.3 detected.
This operating system will use Easy Install. (What's this?)
I will install the operating system later.
The virtual machine will be created with a blank hard disk.
Help
< Back
Next >
Cancel


VMware Workstation - e.x.p build-16540321
WORKSTATION 20H2
New Virtual Machine Wizard
Easy Install Information
This is used to install Ubuntu 64-bit.
Personalize Linux
Full name:
eswar
User name:
eswar_yadav
Password:
Confirm:
Help
< Back
Next >
Cancel



Result

Thus the Virtual Machine was successfully created using **VMware Workstation**, a Type-2 Hypervisor, with the specified CPU, RAM, and storage configuration.

Ex. No: 3 Creating and Testing a VM Snapshot

Aim

To demonstrate virtualization by creating a Snapshot of a Virtual Machine using VMware Workstation and testing it by loading the previous version of the Virtual Machine.

PROCEDURE

Step 1: Open VMware Workstation.

Step 2: Select the previously created Virtual Machine.

Step 3: Power on the Virtual Machine.

Step 4: Make some changes inside the Virtual Machine.

Step 5: Select VM → Snapshot → Take Snapshot.

Step 6: Enter a suitable snapshot name.

Step 7: Click Take Snapshot to save the current state of the Virtual Machine.

Step 8: Make some additional changes to the Virtual Machine**.**

Step 9: Select VM → Snapshot → Snapshot Manager.

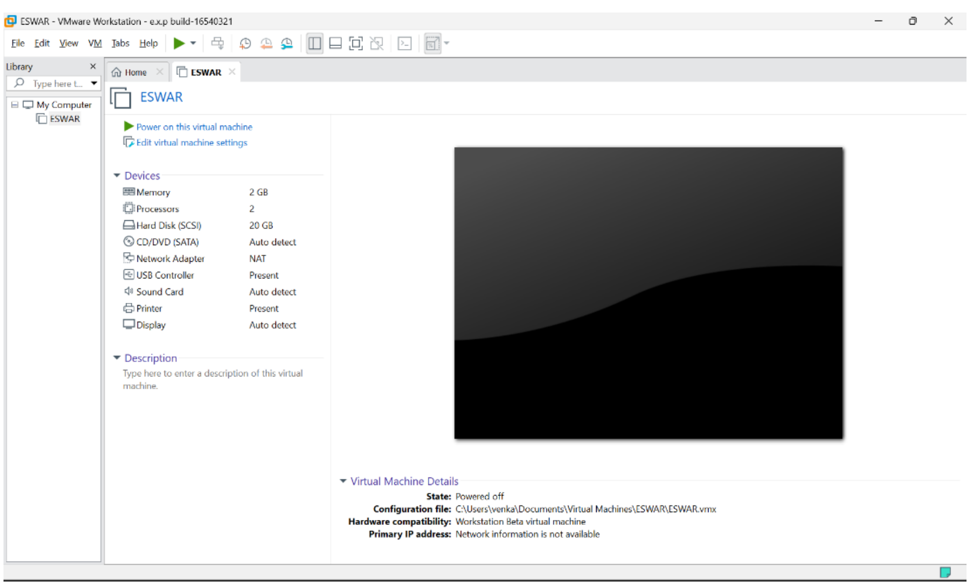
Step 10: Select the previously created snapshot.

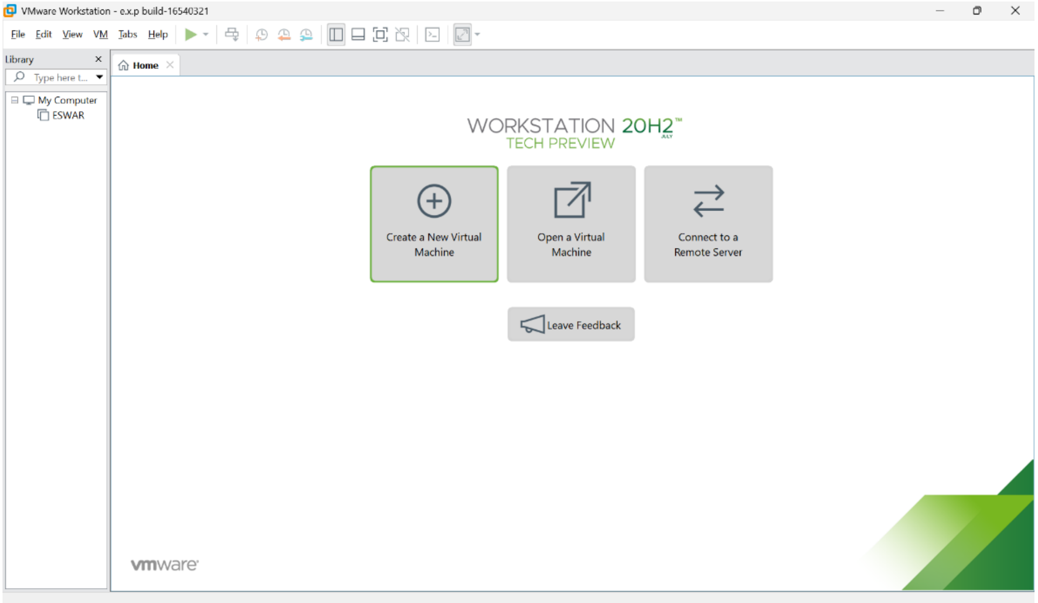
Step 11: Click Go To to restore the Virtual Machine to the saved snapshot.

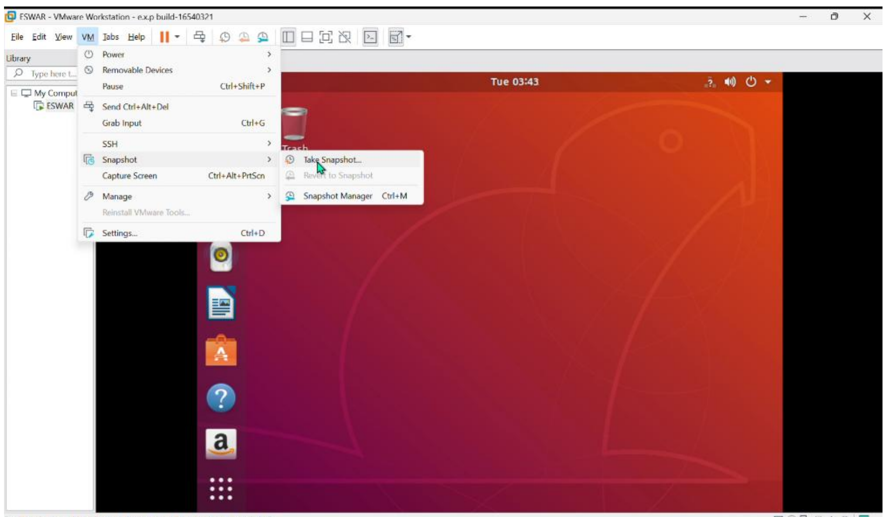
Step 12: Verify that the Virtual Machine has returned to its previous state.

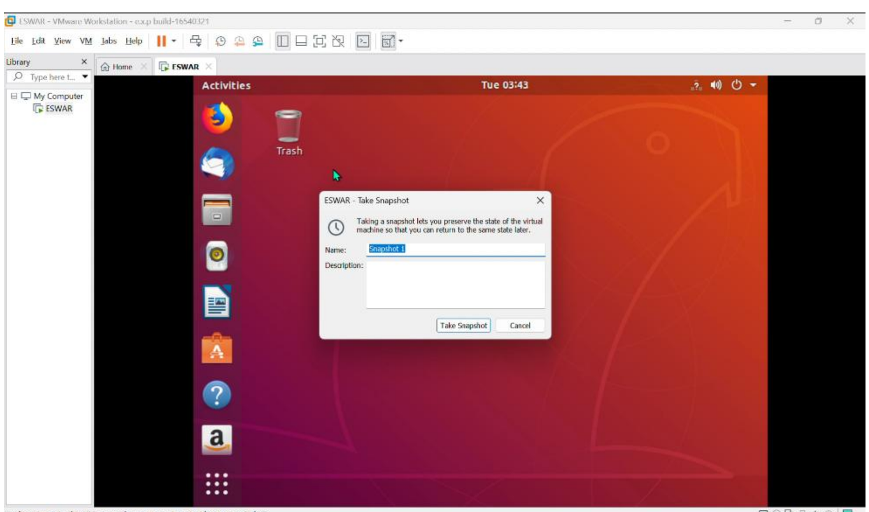
Step 13: The Snapshot has been successfully created and tested.

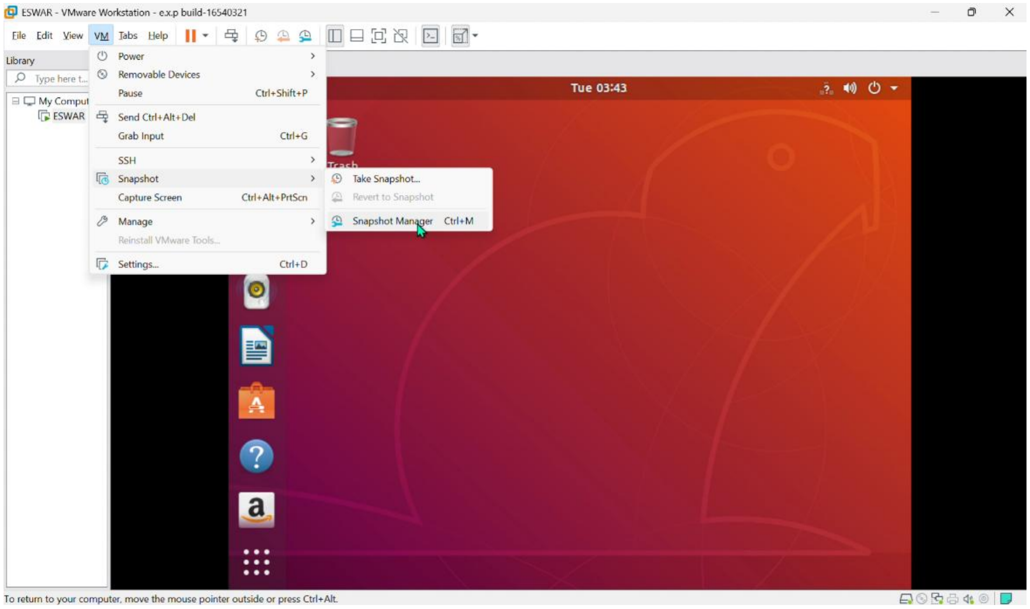
Output

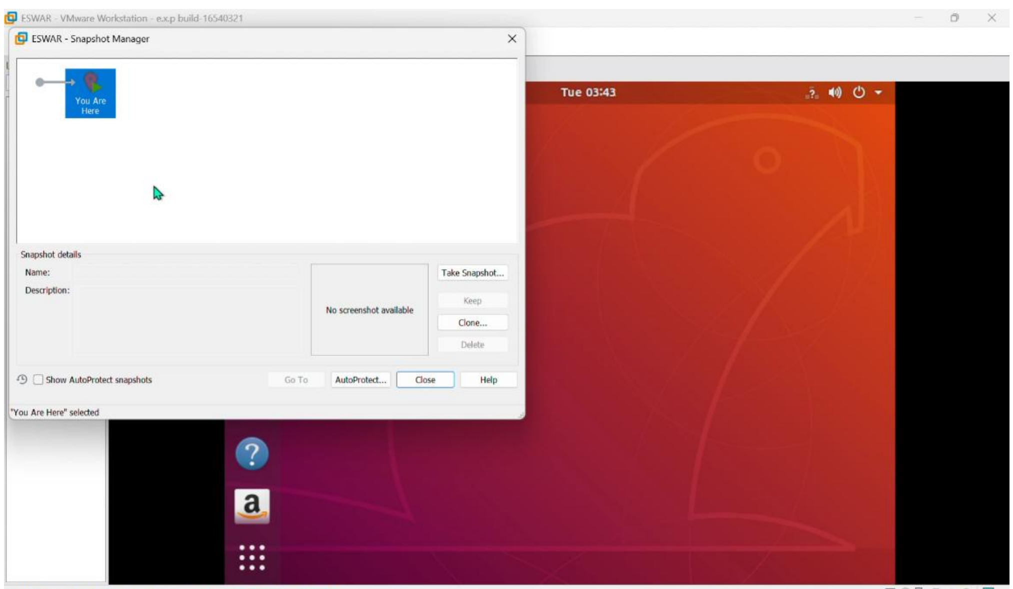



VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Library
Type here t...
My Computer
ESWAR
Home
WORKSTATION 20H2
TECH PREVIEW
Create a New Virtual Machine
Open a Virtual Machine
Connect to a Remote Server
Leave Feedback
vmware


ESWAR - VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Power
Removable Devices
Pause Ctrl+Shift+P
Send Ctrl+Alt+Del
Grab Input Ctrl+G
SSH
Snapshot
Capture Screen Ctrl+Alt+PrtScn
Manage
Reinstall VMware Tools...
Settings... Ctrl+D
Take Snapshot...
Revert to Snapshot
Snapshot Manager Ctrl+M
Tue 03:43


ESWAR - VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Library
My Computer
ESWAR
Home
Activities
Tue 03:43
Trash
ESWAR - Take Snapshot
Taking a snapshot lets you preserve the state of the virtual machine so that you can return to the same state later.
Name: Snapshot 1
Description:
Take Snapshot
Cancel
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.





Result

Thus the Virtual Machine was successfully created using **VMware Workstation**, a Type-2 Hypervisor, with the specified CPU, RAM, and storage configuration.